Gaussian linear optics

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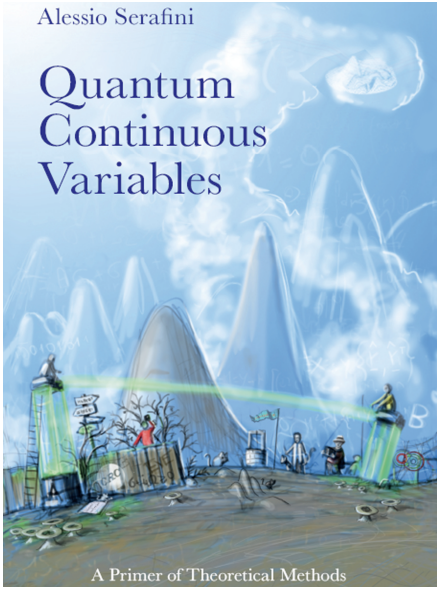
Alessio Serafini
Quantum
Continuous
Variables
A Primer of Theoretical Methods

Collection of harmonic oscillators

- Collection of N harmonic oscillators with quadrature operators $\hat{R} = (\hat{x}_1, \ldots, \hat{x}_N, \hat{p}_1, \ldots, \hat{p}_N)^T$
- $[\hat{x}_i, \hat{p}_j] = 2\mathrm{i}\delta_{ij} \iff [\hat{R}_j, \hat{R}_k] = 2\mathrm{i}\Omega_{jk}$
- Symplectic form $\Omega := \begin{pmatrix} 0_{N\times N} & \mathbb{I}_{N\times N} \\ -\mathbb{I}_{N\times N} & 0_{N\times N} \end{pmatrix}$

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- Phase space $M = T^*\mathbb{R}^N$ has global coordinates $r = (x_1, \ldots, x_N, p_1, \ldots, p_N)$
- Symplectic form: closed nondegenerate 2-form
- $\omega = \sum_{i=1}^N \mathrm{d}x^i \wedge \mathrm{d}p^i$, $\mathrm{d}\omega = 0$
- $\omega = \sum_{i,j} \Omega_{ij}\, \mathrm{d}r^i \otimes \mathrm{d}r^j$

Classical mechanics

- H Hamiltonian, X_H Hamiltonian vector field gives flow lines of constant energy $0 = X_H(H) = \sum_i \left((X_H)_{x_i} \frac{\partial H}{\partial x_i} + (X_H)_{p_i} \frac{\partial H}{\partial p_i} \right)$

$$\implies X_H = \sum_i \left(\frac{\partial H}{\partial p_i} \frac{\partial}{\partial x_i} - \frac{\partial H}{\partial x_i} \frac{\partial}{\partial p_i} \right) \quad \text{i.e. } \dot{x}_i = \frac{\partial H}{\partial p_i}, \dot{p}_i = -\frac{\partial H}{\partial x_i}$$

- Liouville's theorem: $\mathcal{L}_{X_H} \omega = 0$

- Corollary: $\mathcal{L}_{X_H} \omega^N = 0$ (phase space volume is conserved)

First and second moments

Fix a state ρ

Definition (Displacement vector)

$$R_j := \mathrm{Tr}\left[\hat{R}_j \rho\right]$$

Definition (Convariance matrix)

$$\sigma_{ij} := \frac{1}{2}\,\mathrm{Tr}\left[\{\hat{R}_i, \hat{R}_j\}\rho\right] - R_i R_j$$

Energy and photon number

- The harmonic oscillator Hamiltonian is $H = \hat{R}^T \hat{R} = \sum_i (\hat{x}_i^2 + \hat{p}_i^2)$
- Energy $E = \mathrm{Tr}\left[\hat{R}^T \hat{R} \rho\right] = \mathrm{Tr}\, \sigma + R^T R$
- Ladder operators $\hat{x}_i = a_i + a_i^\dagger$, $\hat{p}_i = \mathrm{i}(a_i^\dagger - a_i)$
- Photon number is $\sum_i a_i^\dagger a_i = H/4 - N/2$

Uncertainty principle

Theorem

σ is a valid covariance matrix if it is symmetric, positive, and satisfies $\sigma + \mathrm{i}\Omega \geq 0$.

Proof.

$\delta\hat{R}_i := \hat{R}_i - R_i$

$$\tau_{ij} := \mathrm{Tr}[\rho\delta\hat{R}_i\delta\hat{R}_j] = \mathrm{Tr}\left[\frac{1}{2}\{\delta\hat{R}_i, \delta\hat{R}_j\}\rho + \frac{1}{2}[\delta\hat{R}_i, \delta\hat{R}_j]\rho\right] = \sigma_{ij} + \mathrm{i}\Omega_{ij}$$

$$0 \leq \mathrm{Tr}\left[\rho(y^\dagger\delta\hat{R})(y^\dagger\delta\hat{R})^\dagger\right] = y^\dagger\tau y$$

□

Gaussian states

Definition (Gaussian state)

A Gaussian state is any thermal state $\rho \propto e^{-\beta \hat{H}}$ of a positive Hamiltonian H that is quadratic in the quadrature operators $\hat{x}_i$ and $\hat{p}_i$ (or alternatively, a_i and $a_i^\dagger$). ρ is pure in the limit that $\beta \to \infty$.

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- H is quadratic $\implies$ ρ is defined by first and second quadrature moments R and σ
- Pure state is Gaussian iff $|\psi\rangle = \exp\left[\mathrm{i}\left(\hat{R}^T A \hat{R} + \hat{R}^T b\right)\right]|0\rangle$ for $A \in \mathbb{R}^{2N \times 2N}$ and $b \in \mathbb{R}^{2N}$
- Mixed Gaussian state is partial trace of pure Gaussian state, corresponds to taking the relevant submatrix of σ

Gaussian operations

- BCH $\implies \exp\left[\mathrm{i}\left(\hat{R}^T A \hat{R} + \hat{R}^T b\right)\right] = \exp\left[\mathrm{i}\hat{R}^T A \hat{R}\right] \exp\left[\mathrm{i}\hat{R}^T M b\right]$ for some matrix M
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- $\exp\left[\mathrm{i}\hat{R}^T Mb\right]$ is tensor product of single mode operations, affects first moments R — operations isomorphic to $(\mathbb{R}^{2N}, +)$
- In general, a Gaussian unitary $U\colon \hat{R}_i \mapsto U^\dagger \hat{R}_i U = (S\hat{R} + d)_i$ (e.g. $a \mapsto \mathcal{A}a + \mathcal{B}a^\dagger + \mathcal{C}$)

$$2\mathrm{i}\Omega_{ij} = \left[(S\hat{R} + d)_i, (S\hat{R} + d)_j\right] = \left[S_{ik}\hat{R}_k, S_{jl}\hat{R}_l\right] = 2\mathrm{i}(S\Omega S^T)_{ij}$$

- Hence Gaussian unitaries defined by S and d, where $d \in \mathbb{R}^{2N}$ and $\Omega = S\Omega S^T$

Symplectic group

Definition (Symplectic group)

The *symplectic group* is the group of transformations that preserve the symplectic form; $\mathrm{Sp}(2N, \mathbb{R}) = \mathrm{Sp}(2N) = \{S \in \mathrm{GL}(2N, \mathbb{R}) \mid S\Omega S^T = \Omega\}$.

$$1 = \det \Omega = \det(S\Omega S^T) = (\det S)^2 \implies \det S \neq 0$$
$$\Omega = S^{-1} S\Omega S^T S^{-1\ T} = S^{-1}\Omega S^{-1\ T} \implies S^{-1} \in \mathrm{Sp}(2N)$$

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Theorem

$Sp(2N)$ is not compact

Proof.

$$A_a = \begin{pmatrix} a\mathbb{I}_{N\times N} & 0 \\ 0 & \frac{1}{a}\mathbb{I}_{N\times N} \end{pmatrix} \in \mathrm{Sp}(2N),\ d(A_a, A_b)^2 = n(a-b)^2 + n(1/a - 1/b)^2 \geq (a-b)^2 \qquad \square$$

Passive Gaussian unitaries

From $\hat{R}_i \mapsto (S\hat{R} + d)_i$, a Gaussian unitary acts as $R \mapsto SR + d$, $\sigma \mapsto S\sigma S^T$

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A passive Gaussian unitary is a Gaussian unitary that preserves the total energy (or equivalently total particle number)

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Theorem

A passive Gaussian unitary has $d = 0$ and $S \in \mathrm{K}(N) := \mathrm{Sp}(2N) \cap \mathrm{O}(2N)$

Proof.

$$E = \mathrm{Tr}\,\sigma + \|R\|^2 \mapsto \mathrm{Tr}\left[S\sigma S^T\right] + \|R + d\|^2$$

□

Orthogonal symplectic group is compact

- $\mathrm{Sp}(2N) = f^{-1}(\{\Omega\})$ for continuous map $f\colon A \mapsto A\Omega A^T \implies \mathrm{Sp}(2N)$ is closed
- $\mathrm{O}(2N) = f^{-1}(\{\mathbb{I}\})$ for continuous map $f\colon A \mapsto A^T A \implies \mathrm{O}(2N)$ is closed
- Columns of orthogonal matrix are orthonormal $\implies d(A,B)^2 \leq 8N^2$ for $A, B \in \mathrm{O}(2N) \implies \mathrm{O}(2N)$ is bounded

$$\implies \mathrm{K}(N) := \mathrm{Sp}(2N) \cap \mathrm{O}(2N) \text{ is compact} \implies \text{finite Haar measure!}$$

Theorem

$$\mathrm{U}(N) \cong \mathrm{K}(N) \textit{ via } U \mapsto \begin{pmatrix} \Re U & -\Im U \\ \Im U & \Re U \end{pmatrix}$$

Euler decomposition of symplectic matrix

Theorem (Euler decomposition)

Any real symplectic matrix $S \in Sp(2N)$ can be decomposed as $S = ODO'$ where $O, O' \in K(N) = Sp(2N) \cap O(2N)$ and $D = diag(d_1, \ldots, d_N, 1/d_1, \ldots, 1/d_N)$ for $d_i \in \mathbb{R}_{>0}$.

- Vacuum state is Gaussian with covariance matrix $\mathbb{I}_{2N \times 2N}$
- Squeezed state on one mode is $\mathrm{diag}(\mathrm{e}^{-2r}, \mathrm{e}^{2r})$
- Hence, any pure Gaussian state is $\sigma = (ODO')(ODO')^T = OD^2O^T$

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- Hence, any pure Gaussian state is $\sigma = (ODO')(ODO')^T = OD^2O^T$
- Corresponds to initial product state of squeezed modes acted on by passive Gaussian unitary

Williamson's decomposition

Theorem

Let σ be a positive-definite symmetric $2N \times 2N$ matrix. Then

- $\exists S \in Sp(2N)$ *such that* $S\sigma S^T = D$ *with* $D = \begin{pmatrix} \Lambda & 0 \\ 0 & \Lambda \end{pmatrix}$, *and* Λ *a diagonal matrix of rank* N. *The diagonal entries of* Λ, λ_j, *are called symplectic eigenvalues and are such that* $\pm\lambda_j$ *are eigenvalues of* $\mathrm{i}\Omega\sigma$.
- $\lambda_1, \ldots, \lambda_n$ *does not depend, up to reordering, on the choice of* S *diagonalizing* σ.

Theorem

$$\nu_i = \frac{\lambda_i - 1}{2} \qquad \mathrm{Tr}\,\rho^\alpha = \prod_{i=1}^{N} \frac{1}{(1+\nu_i)^\alpha - \nu_i^\alpha}$$

$$\implies S_\alpha(\rho) = \frac{1}{1-\alpha} \log \mathrm{Tr}\,\rho^\alpha$$

Reyni-2 entropy

- $S_2(\rho) = -\log \mathrm{Tr}\, \rho^2 = \log \prod_{i=1}^{N} \lambda_i = \sum_{i=1}^{N} \log \lambda_i$
- By Williamson's theorem, symplectic eigenvalues turn up twice as $\pm\lambda_j$ in $\mathrm{i}\Omega\sigma$

$$\implies S_2(\rho) = \log \sqrt{|\det(\mathrm{i}\Omega\sigma)|} = \frac{1}{2} \log |\det(\Omega) \det(\sigma)| = \frac{1}{2} \log \det \sigma = \frac{1}{2} \mathrm{Tr} \log \sigma$$

- Entanglement entropy $S_1(\rho) \geq S_2(\rho)$
- arXiv:1203.5116; Reyni-2 entropy satisfies the strong subadditivity inequality for Gaussian states

Gaussian operations

Squeezing operator

The squeezing operator is $\mathcal{S}(r) = \mathrm{e}^{ra^2/2 - ra^{\dagger 2}/2}$, $\hat{x} \mapsto \mathrm{e}^{-r}\hat{x}$, $\hat{p} \mapsto \mathrm{e}^{r}\hat{p}$

$$d = 0, S = \mathrm{diag}(\mathrm{e}^{-r}, \mathrm{e}^{r}) \in \mathrm{Sp}(2), \notin \mathrm{O}(2) \implies \text{active}$$

Phase shift operator

$R(\theta) = \mathrm{e}^{-\mathrm{i}\theta a^\dagger a}$, $a \mapsto \mathrm{e}^{\mathrm{i}\theta} a$. $d = 0, S = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \in \mathrm{K}(1) \implies \text{passive}$

Beamsplitter operator

$B(\beta) = \exp\left(\beta a_1 a_2^\dagger - \beta a_1^\dagger a_2\right)$. $a_1 \mapsto \cos\beta a_1 + \sin\beta a_2, a_2 \mapsto -\sin\beta a_1 + \cos\beta a_2$

$$d = 0, S = \begin{pmatrix} \cos\beta \mathbb{I}_{2\times 2} & \sin\beta \mathbb{I}_{2\times 2} \\ \sin\beta \mathbb{I}_{2\times 2} & \cos\beta \mathbb{I}_{2\times 2} \end{pmatrix} \in \mathrm{K}(2) \implies \text{passive}$$